Faculty of Computing

Universiti Teknologi Malaysia

**Data Structure & Algorithm**

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# Lab Exercise 02

## Instruction

For each algorithm below:

- Count the number of steps executed (loop runs, comparisons, print statements, etc.).
- Express it as a formula in terms of `n`.
- Simplify the formula.
- Write the time complexity in Big-O notation.
- Write down the analysis using the table (same as in the examples in the slide):

| Algorithm | Number of steps |
|---|---|
| | |
| | |
| Total steps | |
| Complexity time | |

## Part 1 (Algorithm 1):

```
1   int x = 1;
2   while (x < n) {
3       cout << "Step " << x << endl;
4       x = x * 2;
5   }
```

## Part 2 (Algorithm 2)

```
1   void sample1(int n) {
2       int sum = 0;
3       for (int i = 1; i <= n; i++)
4           sum += i;
5       cout << sum;
6   }
```

**Part 3 (Algorithm 3):**

```
1   for (int i = 1; i <= n; i = i * 2) {
2       for (int j = 1; j <= n; j++) {
3           cout << i << "," << j << endl;
4       }
5   }
```

**Part 4 (Algorithm 4):**

```
1   int sum = 0;
2   for (int i = 1; i <= n; i++) {
3       for (int j = 1; j <= n; j += i) {
4           sum += j;
5           cout << sum << endl;
6       }
7   }
```

**Part 5 (Algorithm 5):**

```
1   int counter = 0;
2   for (int i = 1; i <= n; i++) {
3       for (int j = 1; j <= n - i; j++) {
4           for (int k = 2; k <= n; k++) {
5               counter++;
6           }
7       }
8   }
```

**Part 6 (Algorithm 6):**

```
1   int count = 0;
2   int limit = 2;
3   int i = 1;
4
5   while (i < n) {
6       limit = limit * 2;
7       i++;
8   }
9
10  for (int j = 1; j <= limit; j++) {
11      count++;
12  }
```